

NCP-AIN AI Fabric Pro Mega Lab: Spectrum CLI Translation Sheet

Purpose

This document translates the current EOS intent for **LEAF1-4** and **BL1-2** into NVIDIA-style Ethernet switch configuration sections. It is intentionally NOS-neutral because exact commands differ across **Cumulus/NVUE**, **Onyx**, and **SONiC on NVIDIA**.

Common translation blocks

- Global routing and system: hostname, IPv4 forwarding, IPv6 forwarding, anycast gateway MAC.
- VLAN database: create VLAN 110, 120, 130, 210, and 4094.
- VRFs: AI_FABRIC, AI_SERVICES, TENANT_B.
- Loopbacks: Lo0 router-id, Lo1 VTEP source, Lo2 shared virtual VTEP, and Lo100 on border nodes.
- Routed uplinks: Ethernet1 and Ethernet2, /31 addressing, MTU 9214, underlay routing.
- MLAG domain: peer-link on Ethernet10-11, peer SVI on VLAN4094, peer IPs preserved.
- Host bonds: Po10, Po11, Po20, Po21, Po30 with the same VLAN intent.
- SVIs and anycast gateways: same IPv4 and IPv6 gateway addresses as EOS.
- VXLAN and EVPN: same VLAN-to-VNI and VRF-to-L3VNI mappings.
- BGP EVPN: same AS 65000, same RR neighbors, same RD and RT intent.
- Border eBGP: only on BL1 and BL2 via Ethernet30.
- QoS and RoCE: attach PFC, ECN, DSCP, and MTU policy at host and uplink interfaces.

EOS to NVIDIA-style section mapping

EOS global routing -> NVIDIA system / routing / anycast gateway sections

EOS VLAN definitions -> NVIDIA bridge VLAN / VLAN database sections

EOS vrf instance -> NVIDIA VRF definition / network-instance sections

EOS Loopback0-2 -> NVIDIA loopback interfaces and VTEP source objects

EOS Ethernet1-2 routed uplinks -> NVIDIA routed ethernet interfaces with OSPF and MTU

EOS Port-Channel100 + Vlan4094 + mlag -> NVIDIA MLAG or MC-LAG peer-link and peer SVI

EOS Port-Channel10/11/20/21/30 -> NVIDIA host bond or LAG interface sections

EOS Vlan110/120/130/210 -> NVIDIA SVI and anycast gateway sections

EOS Vxlan1 -> NVIDIA NVE/VXLAN tunnel and L2/L3 VNI mapping

EOS router ospf -> NVIDIA underlay OSPF section

EOS router bgp evpn -> NVIDIA BGP EVPN and VRF EVPN sections

EOS BL1/BL2 Ethernet30 + ipv4 BGP -> NVIDIA external eBGP interface and AF sections

Node-specific translation guidance

LEAF1

- Keep Ethernet1 10.0.0.0/31 and Ethernet2 10.0.0.12/31 as routed uplinks.
- Peer-link remains Ethernet10-11 to LEAF2 and peer SVI remains VLAN4094 with 10.255.12.1/30.
- Host bond Po10 stays on VLAN110 for GPU-A.
- Host bond Po11 stays on VLAN210 for GPU-C.
- Loopbacks remain Lo0 10.255.0.1, Lo1 10.254.0.1, Lo2 10.254.12.12.

LEAF2

- Keep Ethernet1 10.0.0.2/31 and Ethernet2 10.0.0.14/31.
- Peer SVI remains VLAN4094 with 10.255.12.2/30.
- Use the same MLAG, EVPN, VLAN, VRF, and VNI intent as LEAF1.

LEAF3

- Keep Ethernet1 10.0.0.4/31 and Ethernet2 10.0.0.16/31.
- Peer-link remains Ethernet10-11 to LEAF4 and peer SVI remains VLAN4094 with 10.255.34.1/30.
- Po20 remains VLAN110 for GPU-B.
- Po21 remains VLAN120 for STORAGE.
- Loopbacks remain Lo0 10.255.0.3, Lo1 10.254.0.3, Lo2 10.254.34.34.

LEAF4

- Keep Ethernet1 10.0.0.6/31 and Ethernet2 10.0.0.18/31.
- Peer SVI remains VLAN4094 with 10.255.34.2/30.
- Use the same MLAG, EVPN, VLAN, VRF, and VNI intent as LEAF3.

BL1

- Keep Ethernet1 10.0.0.8/31 and Ethernet2 10.0.0.20/31.
- Keep Ethernet30 198.51.100.0/31 as the routed external edge to PE1.
- Peer-link remains Ethernet10-11 to BL2 and peer SVI remains VLAN4094 with 10.255.78.1/30.
- Po30 remains VLAN130 for COLLECTOR.
- Loopbacks remain Lo0 10.255.0.11, Lo1 10.254.0.11, Lo2 10.254.78.78, Lo100 100.64.10.11.
- Keep external eBGP neighbor 198.51.100.1 remote-as 65301 and advertise 100.64.10.11/32.

BL2

- Keep Ethernet1 10.0.0.10/31 and Ethernet2 10.0.0.22/31.
- Keep Ethernet30 198.51.100.2/31 as the routed external edge to PE2.
- Peer SVI remains VLAN4094 with 10.255.78.2/30.
- Keep loopback and border EVPN intent symmetrical with BL1.

Recommended implementation order

1. hostname, IPv4/IPv6 forwarding, anycast gateway MAC
2. VLANs and VRFs
3. loopbacks

4. routed uplinks with MTU
5. MLAG peer-link and peer SVI
6. host or border LAG interfaces
7. tenant SVIs and anycast gateway
8. VXLAN and EVPN VNI/L3VNI objects
9. OSPF underlay
10. BGP EVPN
11. external eBGP on BL1 and BL2
12. QoS, PFC, and ECN policy

Do not change these values during translation

Keep all /31 links, loopback addresses, VLAN IDs, VNI numbers, VRF names, gateway IPs, RR neighbors, and external BGP peer IPs identical to the EOS lab. That preserves workbook validation and troubleshooting flows.